A

PROJECT REPORT ON

"Smart CardioCare - Heart Disease Prediction Using Machine Learning"

Submitted in

the partial fulfilment of the requirement for the award of the degree

of

BACHELOR OF COMPUTER APPLICATION

from

HEMACHAND YADAV VISHWAVIDYALAYA, DURG (C.G.)

SESSION-2025-26

Project Guided By

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SUBMITTED TO

Rungta College of Science & Technology Ganjpara, Durg

491001 (C.G)

CERTIFICATE OF APPROVAL

This is to certify that the Project work entitled "***Smart CardioCare - Heart Disease Prediction Using Machine Learning***"is carried out under the guidance of **Dr. Ashish Tamarakar**. A student of BCA-III year at **Rungta College of Science & Technology (RCST)** is hereby approved as a credible work in the discipline of Computer Science and Application for the award of degree of **Bachelor of Computer Application** during the year 2025-2026 from **Hemchand Yadav Vishwavidyalaya (C.G.)**.

Head of Department

Dr. Ashish Tamrakar

CERTIFICATE

This is to certify that the Project work entitled " **Smart CardioCare - Heart Disease Prediction Using Machine Learning** " submitted to the **Rungta College of Science & Technology (RCST)** by **Ms. Mahi Wasnik**, Roll No.*53380789*, in partial fulfilment for the requirements relating to nature and standard of award of **Bachelor of Computer Application** degree by, **Hemchand Yadav Vishwavidyalaya, Durg (C.G.)** for the academic year 2025-2026. This project work has been carried out under my guidance.

_____

Dr. Ashish Tamarakar

(Associate Professor)

CERTIFICATE OF EVALUATION

This is to certify that the Project work entitled "**Smart CardioCare – Heart Disease Prediction Using Machine Learning**" is carried out by **Ms. Mahi Wasnik**, a student of BCA–3rd year at **Rungta College of Science & Technology(RCST)** after proper evaluation and examination, is hereby approved as a credible work in the discipline of Computer Science and Application and is done in a satisfactory manner for its Application during the year 2025-2026 from **Hemchand Yadav Vishwavidyalaya, Durg(C.G.)**.

_____ _____

Internal Examiner External Examiner

Date: Date:

DECLARATION

This is to certify that the Project work entitled " Smart CardioCare – Heart Disease Prediction Using Machine Learning " which is submitted by me in partial fulfilment for the award of degree of Bachelor of Computer Application at Rungta College of Science & Technology (RCST), comprises the original work carried out by me. I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full for the award of any other degree or diploma in this Institute or any other Institute or University.

Place: Durg, C.G. Mahi Wasnik

Date: Roll No: 53380789

ACKNOWLEDGEMENT

I would like to express my special thanks of gratitude to my teacher Dr. Ashish Tamarakar who gave me the golden opportunity to build this wonderful software named "Smart CardioCare – Heart Disease Prediction Using Machine Learning", which also helped me in doing a lot of research and I came to know about so many new things. I am really thankful to him.

I would also like to thanks to the faculty members who provided me the resources such as Internet connection, Computer System, Working environment etc.

My sincere thanks go to my friends who offered their cooperation, motivation, and assistance whenever needed. Their encouragement kept me focused and determined. Finally, I extend my deepest appreciation to my family for their constant support, patience, and moral encouragement during this journey.

Project Synopsis

Project Title:

Smart CardioCare: Heart Disease Prediction Using Machine Learning

Synopsis:

- **Purpose:** Develop a web-based system that predicts the risk of heart disease using machine learning, enabling early detection and preventive care.
- **User Input:** Users can enter personal health parameters such as age, blood pressure, cholesterol, and other relevant medical data.
- **Prediction:** The system employs a trained machine learning model using Scikit-learn to assess the likelihood of heart disease accurately.
- **Output:** Provides instant results with clear visualizations of risk levels for easy understanding.
- **Recommendations:** Suggests lifestyle modifications and preventive measures based on the prediction outcome.
- **User Interface:** Designed to be simple, intuitive, and user-friendly, ensuring seamless interaction for all users.
- **Scope:** Serves as a personal heart health monitoring tool and a supplementary resource for healthcare providers.
- **Objective:** Facilitate early detection of heart conditions, promote preventive care, and empower users to make informed health decisions.

Technologies Used:

- Programming Language: Python
- Machine Learning Library: Scikit-learn, NumPy, Pandas
- Database: SQLite
- Web Framework: Django
- Visualization Tool: Matplotlib
- Frontend Technologies: HTML, CSS, JavaScript

Abstract

The Smart CardioCare project is a **machine learning-based heart disease prediction system** designed to help users assess their risk of heart disease quickly and efficiently. With the increasing prevalence of cardiovascular diseases, early detection has become crucial for prevention and timely medical intervention. This system allows users to input essential health parameters such as age, blood pressure, cholesterol, and ECG results, which are then processed by a trained machine learning model to provide accurate risk predictions.

The platform also visualizes the heart disease risk through intuitive charts and offers personalized guidance, including lifestyle suggestions and recommendations for consulting doctors. By combining a **user-friendly web interface**, real-time prediction, and clear insights, Smart CardioCare empowers users to make informed decisions about their health. The system is designed to be accessible, fast, and reliable, ensuring that early detection and prevention of heart disease are within reach for everyone.

Key technologies used: Python, Machine Learning, SQLite database, and web-based development tools.

MAIN REPORT

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1.Introduction

Heart disease is one of the leading causes of death worldwide and continues to pose a serious threat to global health. According to medical research, many cardiovascular conditions can be prevented if detected at an early stage. Early diagnosis and timely medical intervention play a crucial role in reducing complications and improving patient survival rates. However, traditional diagnosis methods may sometimes be time-consuming and require continuous medical supervision.

With advancements in technology, Machine Learning has emerged as a powerful tool in the healthcare sector. Machine Learning algorithms can analyze large amounts of medical data and identify hidden patterns that may not be easily recognized through manual observation. By using patient health parameters, predictive models can assist in estimating the risk of heart disease with considerable accuracy.

Smart CardioCare is a web-based application developed to predict the risk of heart disease using Machine Learning techniques. The system analyzes important health parameters such as age, blood pressure, cholesterol level, heart rate, and other clinical inputs to determine whether a person is at potential risk. Based on these inputs, the model generates a prediction result along with an accuracy percentage to provide better clarity to the user.

The main objective of this project is to assist users in understanding their heart health status and promote awareness about preventive healthcare measures. The system also provides prediction history tracking, downloadable reports, and role-based dashboards for patients, doctors, and administrators. Furthermore, the project demonstrates how intelligent systems can support healthcare decision-making by providing quick, accessible, and data-driven insights.

Overall, Smart CardioCare highlights the practical implementation of Machine Learning in the medical domain and showcases how technology can contribute to early risk detection and improved healthcare management.

2.Objective & Scope of the Project

Objective of the Project:

The primary objective of the Smart CardioCare project is to design and develop an intelligent web-based system that predicts the risk of heart disease using machine learning techniques. The system analyzes important health parameters such as age, blood pressure, cholesterol level, and other medical attributes to determine the likelihood of heart disease.

The project aims to provide users with quick, accurate, and easy-to-understand prediction results. It also promotes awareness by including lifestyle analysis to help individuals understand factors affecting heart health. The overall goal is to support early detection and preventive healthcare through technology.

Scope of the Project:

The scope of the Smart CardioCare system is to offer a user-friendly platform where individuals can input their medical data and receive instant heart disease risk predictions. The system is designed to assist users in monitoring their heart health and taking preventive measures at an early stage.

This project can be used as a supportive tool for healthcare awareness and preliminary screening. In the future, the system can be enhanced by incorporating more medical parameters, improving model accuracy, and integrating real-time healthcare data for more advanced prediction capabilities.

3.Main Pages and Functions of the Web System

The Smart CardioCare web application consists of multiple pages and dashboards designed for different users, including patients, doctors, and administrators. Each module provides specific features to ensure efficient heart disease prediction and management.

1. Home Page:

The home page serves as the entry point of the system. It provides basic information about the application and allows users to navigate to login or registration. It also introduces the purpose of Smart CardioCare in predicting heart disease risk.

2. User (Patient) Dashboard:

The patient dashboard allows users to enter their health details and obtain heart disease predictions using the machine learning model. Patients can view their prediction results, access their prediction history, and monitor their heart health status. The dashboard also allows patients to download their prediction history and submit feedback.

3. Prediction Page:

This page contains a form where users enter medical parameters such as age, blood pressure, cholesterol level, and other health indicators. Based on these inputs, the system analyzes the data and displays the prediction result along with risk level and recommendations.

4. Doctor Dashboard:

The doctor dashboard allows doctors to view patient prediction records and monitor patient health data. Doctors can also view their profile information and access prediction history related to patients for medical reference.

5. Admin Dashboard:

The admin dashboard provides complete control over the system. The administrator can add and manage doctor records, view patient details, monitor prediction records, and review user feedback. This helps in maintaining system organization and ensuring proper system management.

6. Feedback Page:

This page allows patients to submit feedback about the system. The feedback is stored and can be viewed by the administrator for system improvement.

7. Prediction History Page:

This page stores all previous prediction results of the user. Patients and doctors can view prediction history, and patients also have the option to download their prediction records.

8. Smart Cardio Analyzer:

Smart CardioCare includes an intelligent Smart Health Calculator that automatically processes user medical and lifestyle inputs to assess heart disease risk. The system evaluates parameters such as blood pressure, cholesterol, heart rate, and lifestyle habits including exercise, diet, sleep, smoking, and alcohol consumption. It then generates a prediction result along with a lifestyle awareness score. This feature enhances user understanding of heart health and promotes preventive healthcare through personalized insights.

4. Theoretical Background

This project is based on key concepts of machine learning, healthcare prediction, and intelligent decision support systems:

Machine Learning in Healthcare:

Machine learning is widely used in healthcare to analyze patient data and predict diseases at an early stage. It helps identify patterns from medical parameters such as age, blood pressure, cholesterol level, and heart rate. In Smart CardioCare, machine learning is used to predict the risk of heart disease based on user input.

Supervised Learning Algorithms:

This project uses supervised learning algorithms such as Logistic Regression and Gradient Boosting. These algorithms are trained on historical medical datasets with known outcomes, allowing the system to classify whether a person has a low or high risk of heart disease.

Predictive Modeling:

Predictive modeling involves using trained machine learning models to forecast future outcomes. The system analyzes user health data and generates a prediction result with accuracy. This helps in early risk detection and preventive healthcare.

Data Analysis and Processing:

Data analysis is essential for improving model performance. Libraries such as NumPy, Pandas, and Scikit-learn are used to process, clean, and analyze the dataset. Proper data handling ensures reliable and accurate predictions.

Decision Support System:

Smart CardioCare acts as a decision support system by providing prediction results, risk scores, and lifestyle awareness. It helps users understand their heart health condition and supports informed healthcare decisions.

Data Visualization and User Interface:

The system displays prediction results using visual elements such as progress bars and donut charts. These visualizations improve user understanding and make the system interactive and easy to use.

5. Definition of Problem

Heart disease is one of the leading causes of death worldwide, and early detection plays a crucial role in preventing severe health complications. However, many individuals are unaware of their heart health status due to lack of regular medical checkups, limited access to healthcare facilities, or absence of easy-to-use diagnostic tools. Traditional diagnosis methods require clinical tests, expert consultation, and hospital visits, which may not always be convenient, affordable, or accessible for everyone.

Additionally, people often ignore early warning signs such as high blood pressure, cholesterol imbalance, poor lifestyle habits, and irregular heart activity. Without proper monitoring and awareness, these risk factors can develop into serious cardiovascular diseases. There is a need for a system that can analyze health parameters efficiently and provide early prediction and awareness to users.

The main problem addressed in this project is the absence of an intelligent, accessible, and user-friendly platform that can assist in predicting heart disease risk based on individual health data. Existing systems may lack proper visualization, personalized feedback, and awareness features that help users understand their health condition clearly.

To solve this problem, the Smart CardioCare system is designed to use machine learning algorithms to analyze health parameters and predict the risk of heart disease. The system provides prediction results, risk score visualization, lifestyle awareness indicators, and guidance to help users understand and improve their heart health. This helps in early detection, promotes preventive care, and supports better health awareness.

6.System Analysis and Design

System Analysis:

In the Smart CardioCare project, system analysis was performed to understand how heart disease prediction can be automated using machine learning and how users can interact with the system efficiently.

The analysis identified that users need a simple and reliable platform where they can enter their health details and receive an instant prediction of heart disease risk. The system must also provide clear results, risk visualization, and guidance for further action.

The key requirements identified are:

- The system should allow users to input health parameters such as age, blood pressure, cholesterol, and other medical data.
- The system should process this data using a trained machine learning model.
- The system should display prediction results clearly in an understandable format.
- The system should provide recommendations based on the prediction.
- The system should be easy to use, fast, and accessible.

The analysis also ensured that the system is user-friendly, reliable, and capable of handling prediction tasks efficiently.

Existing System:

- Diagnosis is done manually by doctors.
- Requires hospital visits and medical tests.
- No instant prediction system.
- Time-consuming process.
- Limited accessibility.
- No automated machine learning support.

Proposed System:

- Web-based heart disease prediction system.
- Uses machine learning algorithms for prediction.
- Provides instant results.
- User-friendly interface.
- Displays visual charts and AI explanation.
- Helps in early detection and prevention.

System Design:

System Design is the process of defining the architecture, components, and overall structure of the system. It converts the system requirements into a technical solution.

The Smart CardioCare system is designed as a web-based application that allows users to interact through a graphical interface. The design consists of three main components:

1. User Interface Design:

The user interface allows users to enter their health information and view prediction results. The interface is designed to be simple, clean, and easy to understand. It includes:

- Input form for health parameters
- Prediction result display
- Risk score visualization
- Lifestyle awareness chart
- Doctor recommendation section

2. Processing and Prediction Design:

- This component handles the core functionality of the system. It includes:
- Collecting input data from the user
- Sending the data to the machine learning model
- Processing the data using trained algorithms
- Generating prediction results

The machine learning model analyzes the input data and predicts whether the user has a low or high risk of heart disease.

3. Database Design:

The system uses a database to manage application data. The database stores:

- User input information
- Doctor details
- System data required for operation
- SQLite is used as the database due to its simplicity and efficiency.

System Architecture Overview:

The Smart CardioCare system follows a three-layer architecture:

- Presentation Layer – User interface where users interact with the system
- Application Layer – Processes data and performs prediction using machine learning
- Data Layer – Stores and retrieves data from the database

This design ensures that the system is organized, efficient, and easy to maintain.

User Roles and Responsibilities:

The Smart CardioCare system supports multiple user roles to ensure proper system functionality and management.

1. Patient:

The patient can enter health details, receive heart disease prediction results, view risk scores, and monitor their prediction history. The patient can also view recommended doctors and lifestyle awareness information.

2. Doctor:

The doctor can view patient prediction records, analyze patient health risk levels, and monitor patient data. This helps doctors understand patient conditions and provide appropriate medical guidance.

3. Administrator:

The administrator manages the overall system. The admin can add and manage doctor records, view patient details, and monitor system feedback. This ensures proper system control and smooth operation.

This role-based structure improves system organization, security, and usability.

7. Methodology Adopted

The Smart CardioCare project follows a structured methodology to ensure accurate heart disease prediction and an efficient user experience. The key steps are:

Data Collection:

- Heart-related datasets (like patient age, blood pressure, cholesterol, etc.) are gathered from reliable sources.
- Data is cleaned and preprocessed to handle missing values and standardize formats.

Data Analysis:

- Exploratory Data Analysis (EDA) is performed to understand trends, correlations, and patterns in the dataset.
- Important features influencing heart disease risk are identified.

Machine Learning Model Development:

- Algorithms such as Gradient Boosting and Logistic Regression are applied.
- Models are trained on the dataset and evaluated for accuracy, precision, and recall.

Prediction System Implementation:

- The trained model is integrated into the web-based application.
- Users can input their health parameters and receive instant predictions.

Result Visualization:

- Prediction outcomes are displayed clearly using visual charts, risk scores, and recommendations.
- Helps users understand their health status and take preventive measures.

Testing and Validation:

- The system is tested for usability, accuracy, and reliability.
- Feedback from sample users is incorporated for improvements.

8. Hardware and Software Environment

Hardware Requirements:

- Processor: Intel Core i3 or higher
- RAM: 4 GB minimum (8 GB recommended)
- Storage: Minimum 100 MB free space for application and dataset
- Display: 1024 x 768 resolution or higher
- Internet Connection: Required for web application access

Software Requirements:

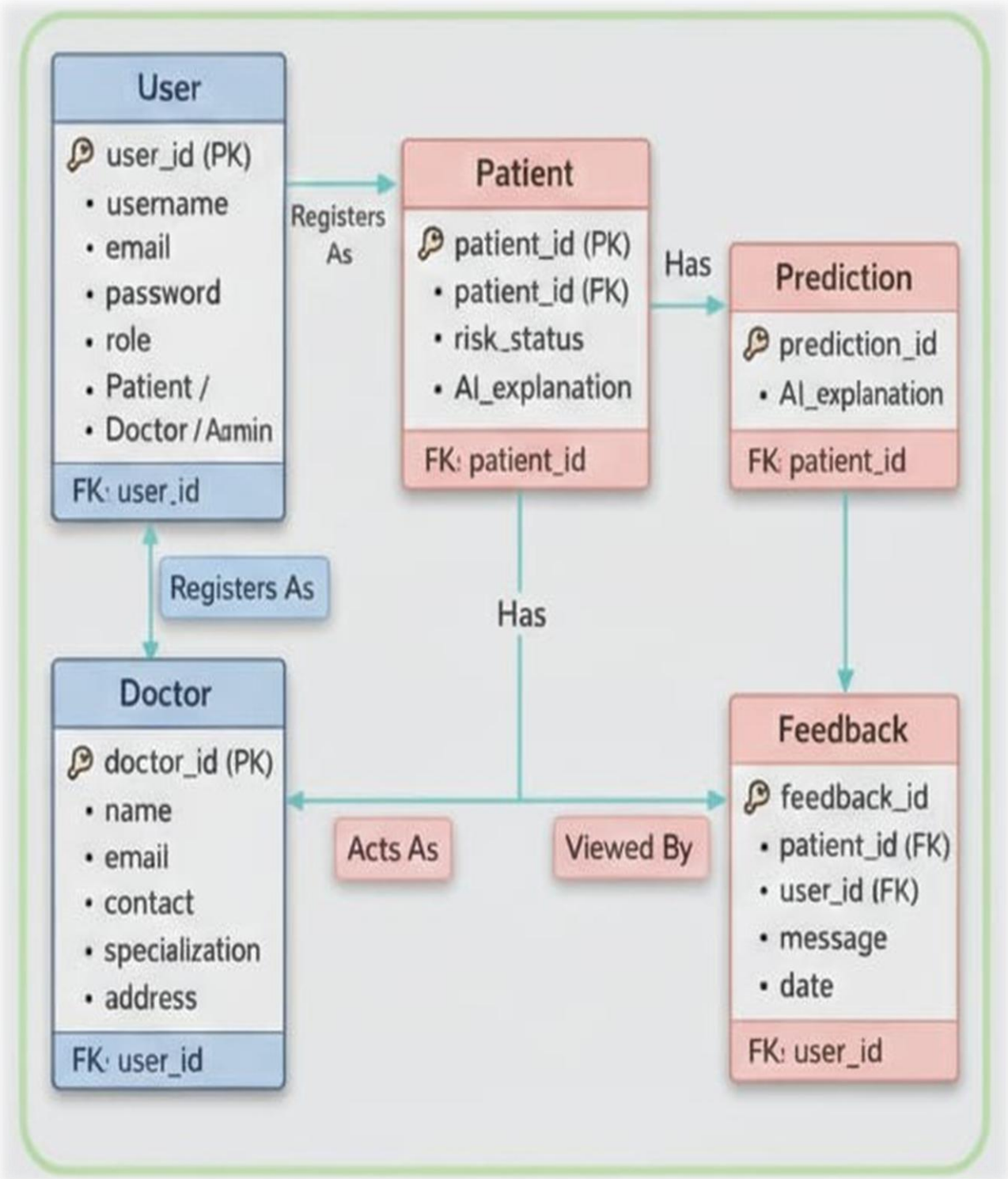
- Operating System: Windows 10/11, macOS, or Linux
- Web Browser: Google Chrome, Mozilla Firefox, or Microsoft Edge
- Programming Languages:
- Python (for machine learning and backend processing)
- HTML, CSS, JavaScript (for frontend development)
- Frameworks and Libraries:
- Django (Web framework for backend)
- Bootstrap (Frontend styling)
- NumPy, Pandas, scikit-learn (Machine Learning and data processing)
- Database: SQLite (for storing user data and system records)

Additional Tools:

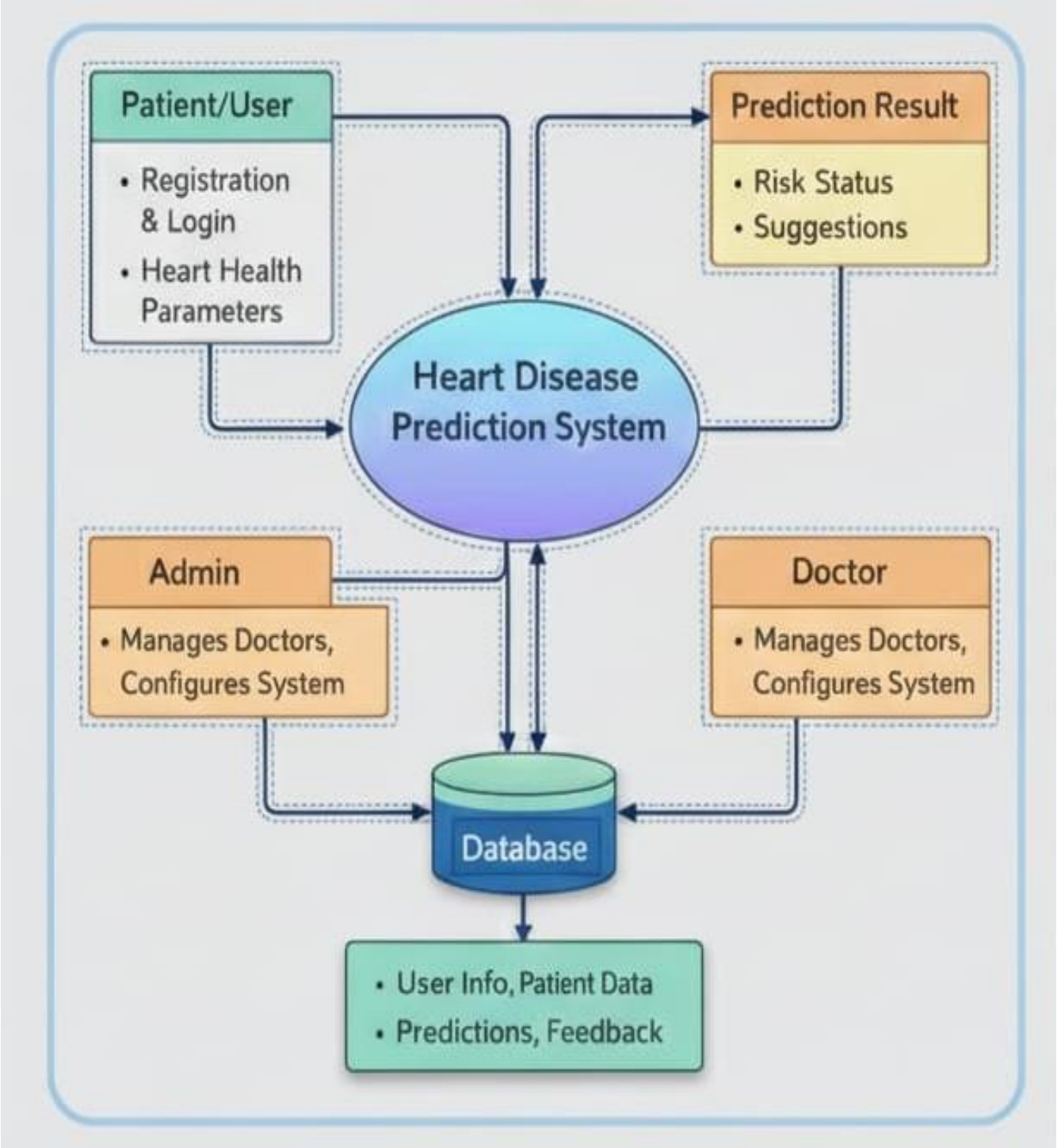
- Code editor (VS Code, PyCharm, or similar)
- Jupyter Notebook (for testing and training machine learning models)

Note: The environment has been configured to ensure smooth execution of the Smart CardioCare system, with efficient model prediction, data management, and user interaction.

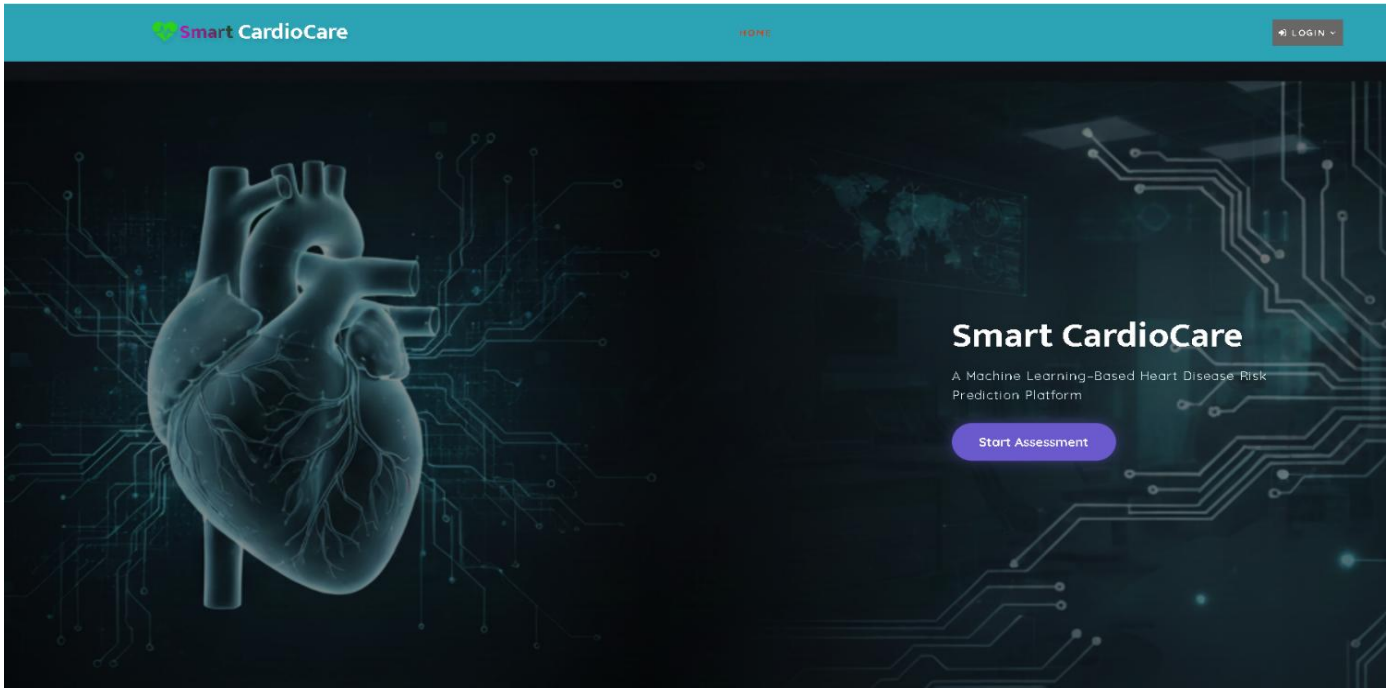
9.ERD(Entity Relationship Diagram)

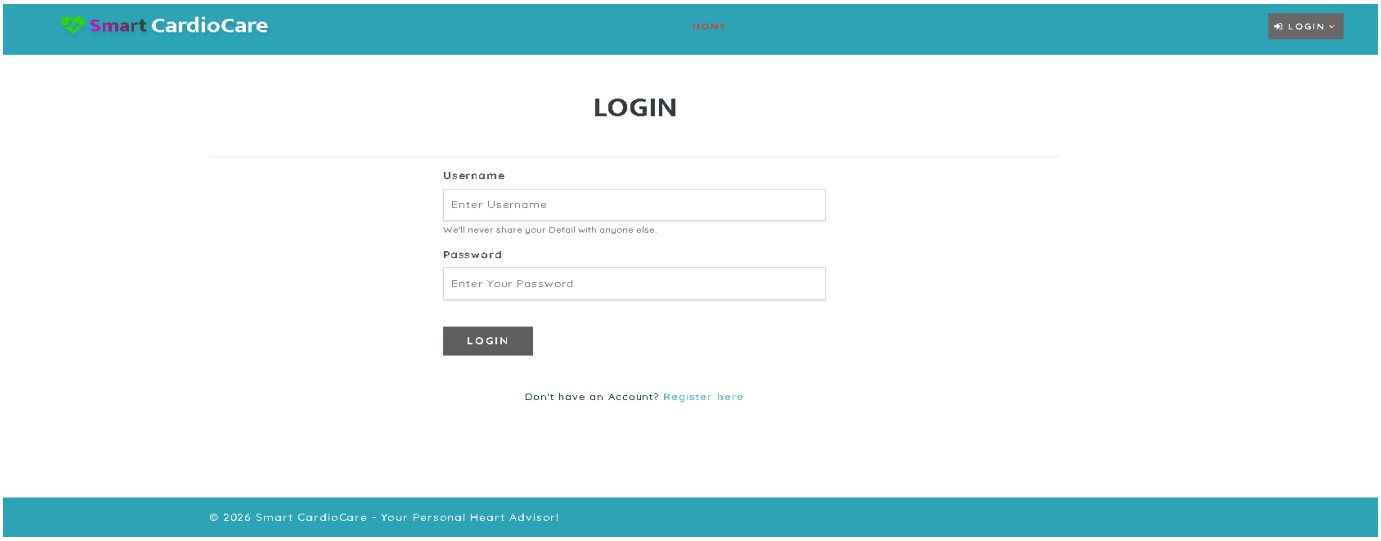


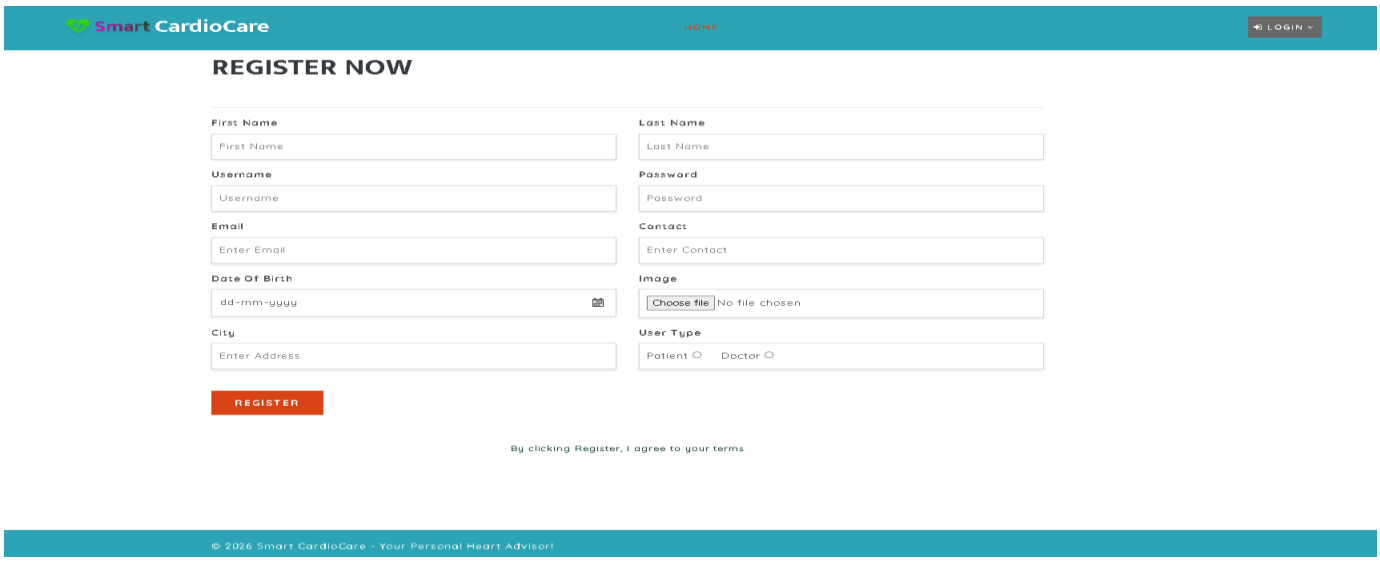
10.DFD(Data Flow Diagram



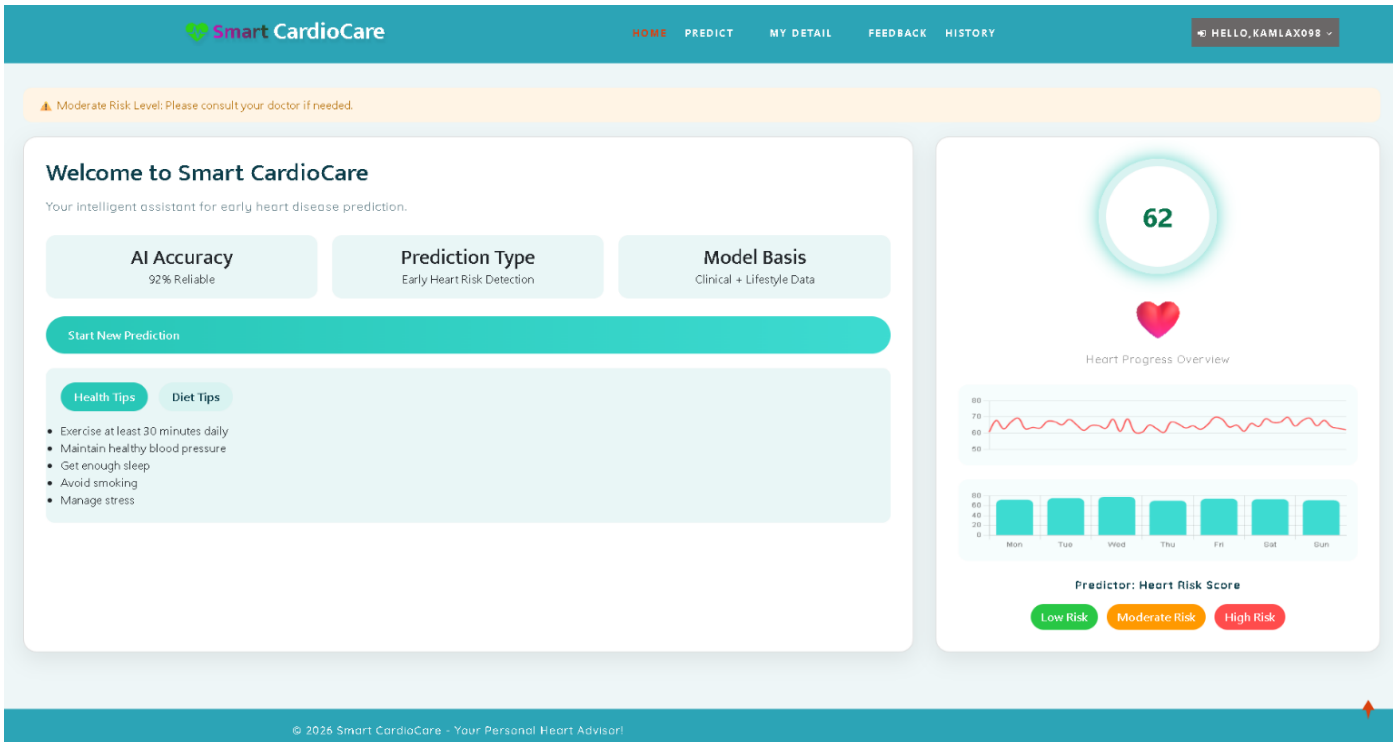
11.Input and Output Screening

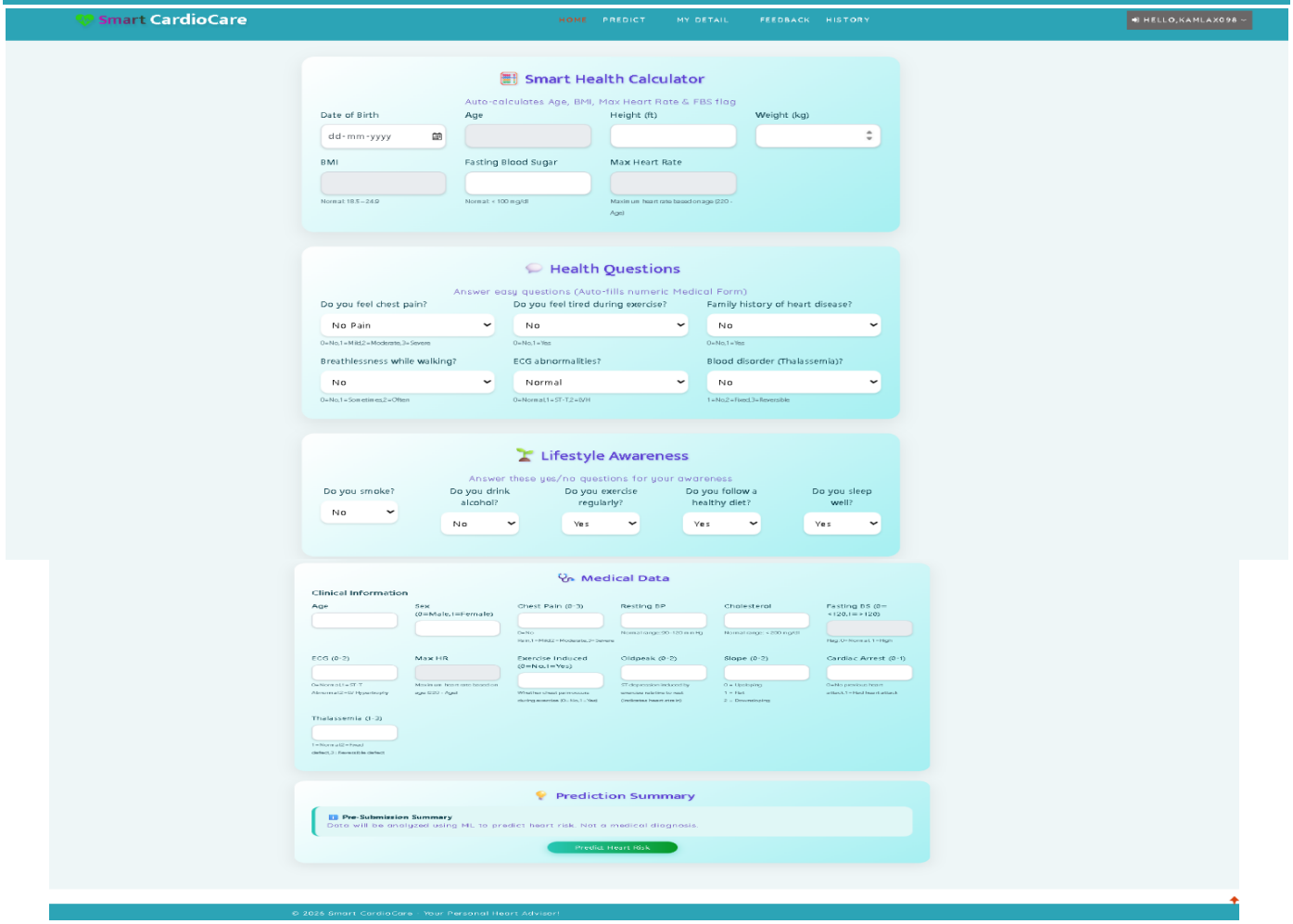






Patient Dashboard-





Prediction Result Page-

AI Prediction Output

Model Accuracy: 88.524590163934 42%

Low Risk: You are healthy. No heart disease detected.

Heart Disease Risk Score

25% Risk

AI Explanation

The machine learning model analyzed your health parameters such as age, blood pressure, cholesterol level, and ECG results to predict heart disease risk.

Next Steps Based on Your Result

These recommendations are personalized based on your heart risk assessment.

Step 1
Maintain your healthy lifestyle habits

Step 2
Continue regular exercise and balanced diet

Step 3
Schedule routine heart checkups annually

This guidance is for awareness only and does not replace professional medical advice.

Lifestyle Awareness

Good lifestyle habits reduce heart risk.

Available Doctors Near You

Saurabh Suman
bhuwanbhaskar761@gmail.com
7876832632
Agra
Contact

Ashok Panjwani
ashok@gmail.com
9229465034
Bhopal
Contact

Vikram Chauhan
mahiwasnik@gmail.com
9244791692
Ahemdabad
Contact

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AI Prediction Output

Model Accuracy: 88.524590163934 42%

High Risk: You may have heart disease symptoms.

Heart Disease Risk Score

80% Risk

AI Explanation

The machine learning model analyzed your health parameters such as age, blood pressure, cholesterol level, and ECG results to predict heart disease risk.

Next Steps Based on Your Result

These recommendations are personalized based on your heart risk assessment.

Step 1
Consult a cardiologist immediately

Step 2
Avoid strenuous activities without medical advice

Step 3
Follow prescribed medication and lifestyle changes strictly

This guidance is for awareness only and does not replace professional medical advice.

Lifestyle Awareness

Good lifestyle habits reduce heart risk.

Available Doctors Near You

Saurabh Suman
bhuwanbhaskar761@gmail.com
7876832632
Agra
Contact

Ashok Panjwani
ashok@gmail.com
9229465034
Bhopal
Contact

Vikram Chauhan
mahiwasnik@gmail.com
9244791692
Ahemdabad
Contact

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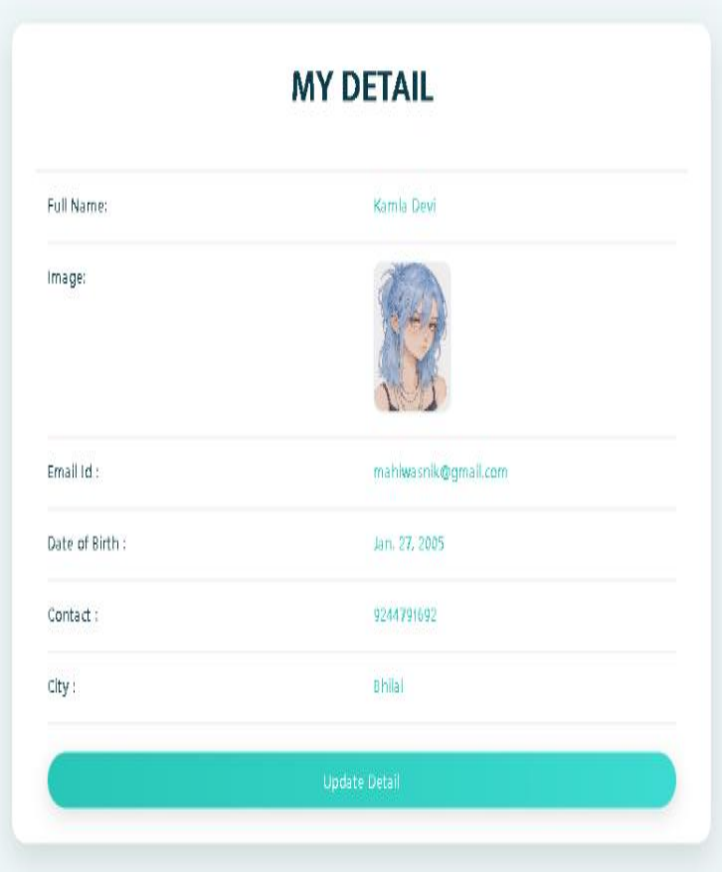
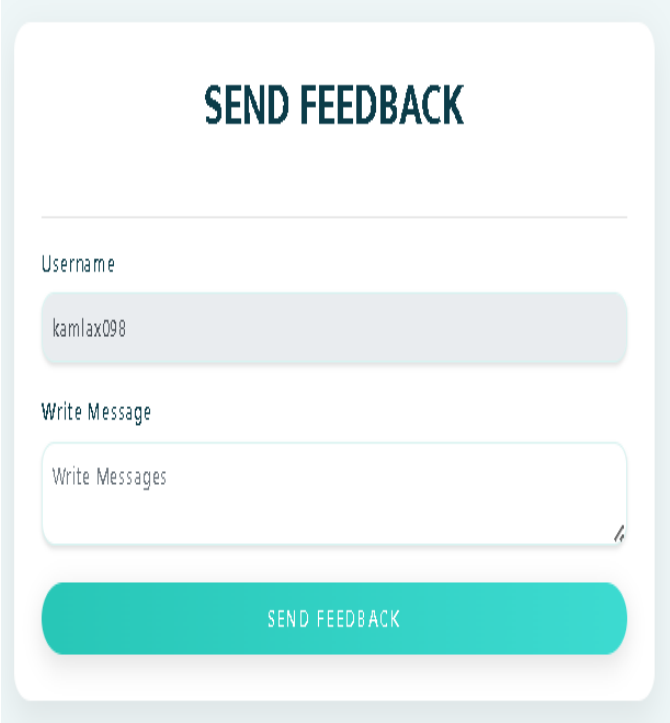
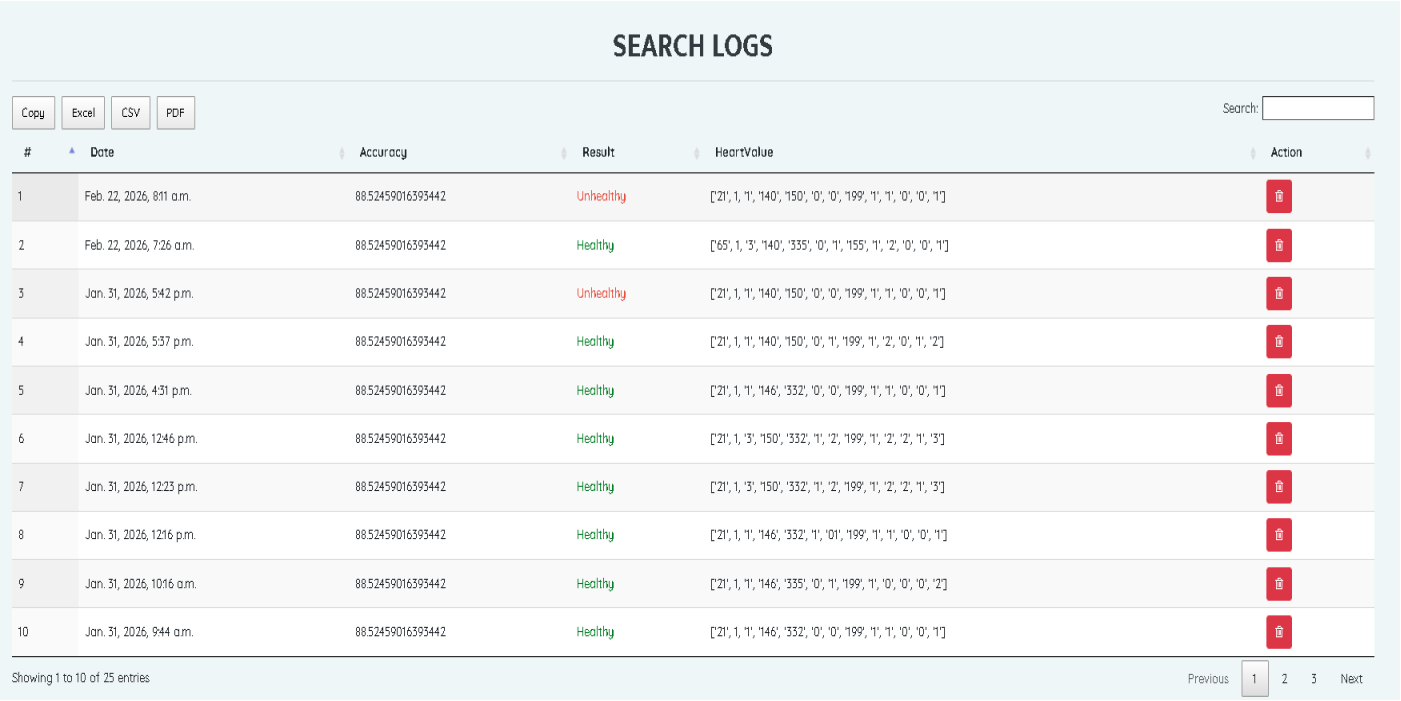
Patient's Profile-





- Patients can send feedback or suggestions through this page, which are stored for the admin to review.

Prediction History-



SEARCH LOGS

Copy Excel CSV PDF

Search:

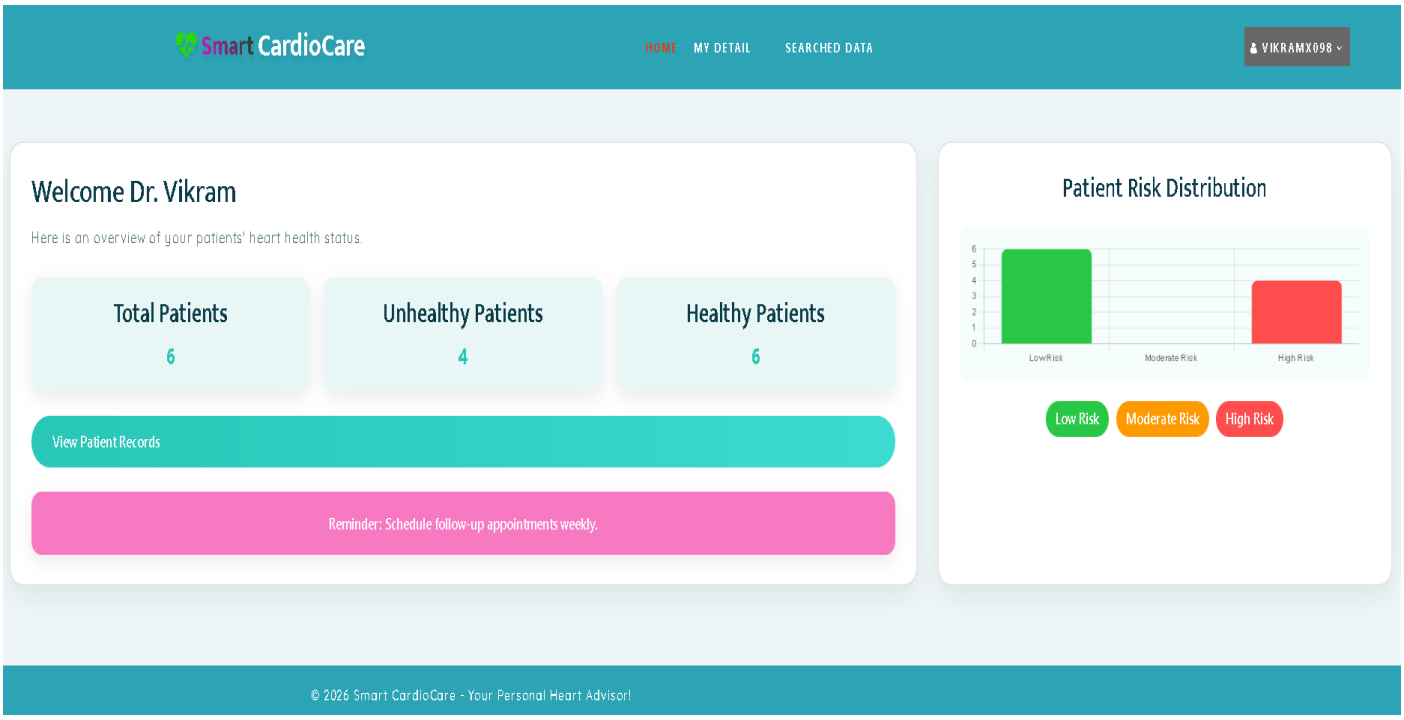
#	Date	Accuracy	Result	HeartValue	Action
1	Feb. 22, 2026, 8:11 a.m.	88.52459016393442	Unhealthy	['21', 1, '1', '140', '150', '0', '0', '199', '1', '1', '0', '0', '1']	
2	Feb. 22, 2026, 7:26 a.m.	88.52459016393442	Healthy	['65', 1, '3', '140', '335', '0', '1', '155', '1', '2', '0', '0', '1']	
3	Jan. 31, 2026, 5:42 p.m.	88.52459016393442	Unhealthy	['21', 1, '1', '140', '150', '0', '0', '199', '1', '1', '0', '0', '1']	
4	Jan. 31, 2026, 5:37 p.m.	88.52459016393442	Healthy	['21', 1, '1', '140', '150', '0', '1', '199', '1', '2', '0', '1', '2']	
5	Jan. 31, 2026, 4:31 p.m.	88.52459016393442	Healthy	['21', 1, '1', '146', '332', '0', '0', '199', '1', '1', '0', '0', '1']	
6	Jan. 31, 2026, 12:46 p.m.	88.52459016393442	Healthy	['21', 1, '3', '150', '332', '1', '2', '199', '1', '2', '2', '1', '3']	
7	Jan. 31, 2026, 12:23 p.m.	88.52459016393442	Healthy	['21', 1, '3', '150', '332', '1', '2', '199', '1', '2', '2', '1', '3']	
8	Jan. 31, 2026, 12:16 p.m.	88.52459016393442	Healthy	['21', 1, '1', '146', '332', '1', '01', '199', '1', '1', '0', '0', '1']	
9	Jan. 31, 2026, 10:16 a.m.	88.52459016393442	Healthy	['21', 1, '1', '146', '335', '0', '1', '199', '1', '0', '0', '0', '2']	
10	Jan. 31, 2026, 9:44 a.m.	88.52459016393442	Healthy	['21', 1, '1', '146', '332', '0', '0', '199', '1', '1', '0', '0', '1']	

Showing 1 to 10 of 25 entries

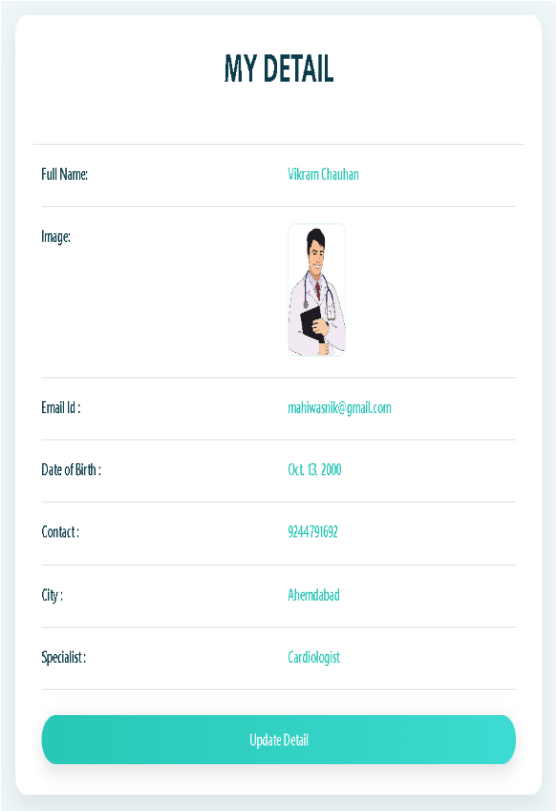
Previous 1 2 3 Next

- The Patient can download their Prediction History.

Doctor Dashboard-

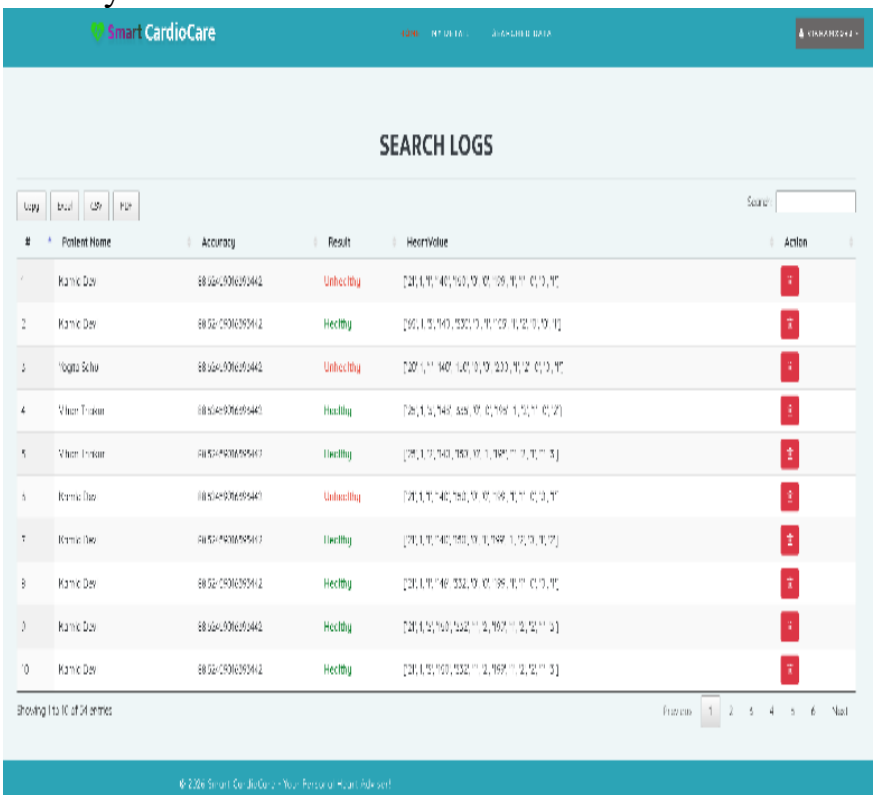


Doctor's Profile-

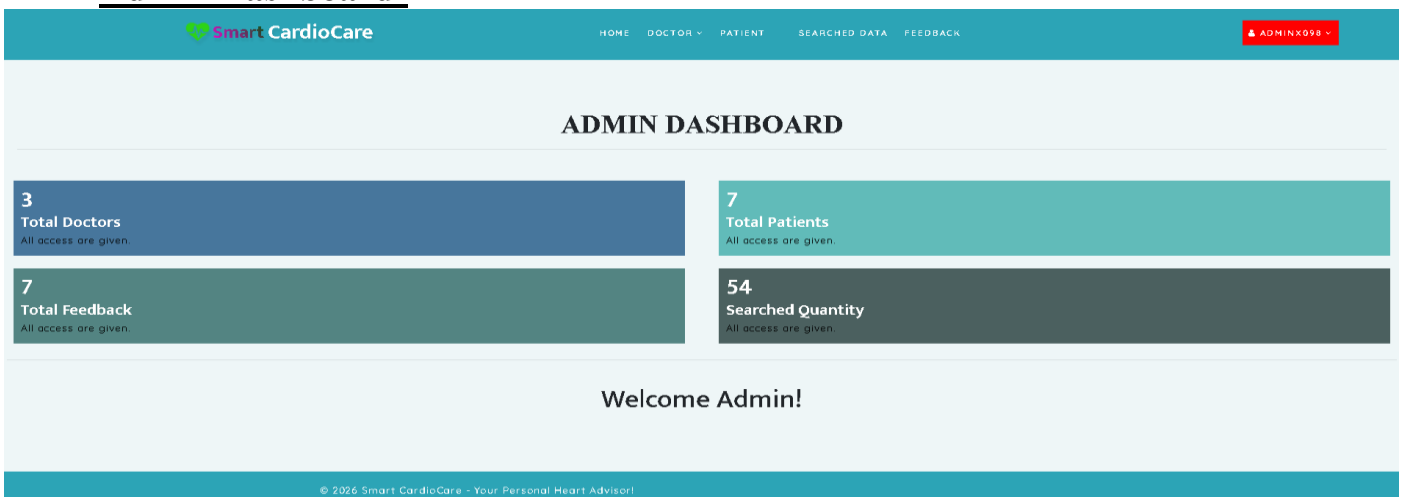


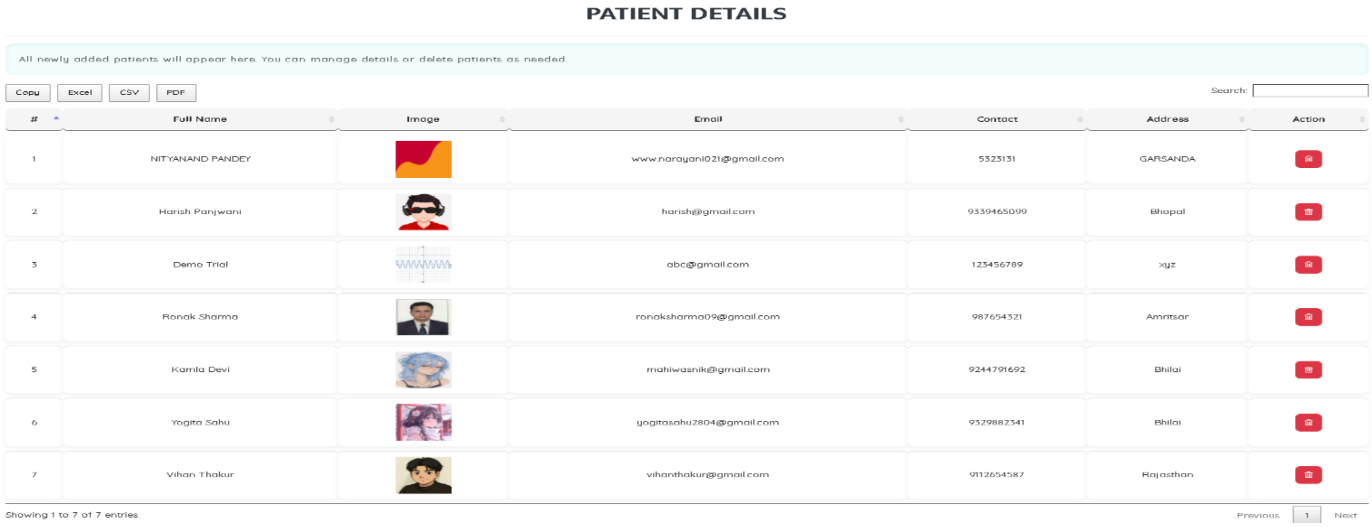
- The Doctor can view and update their personal and professional details.

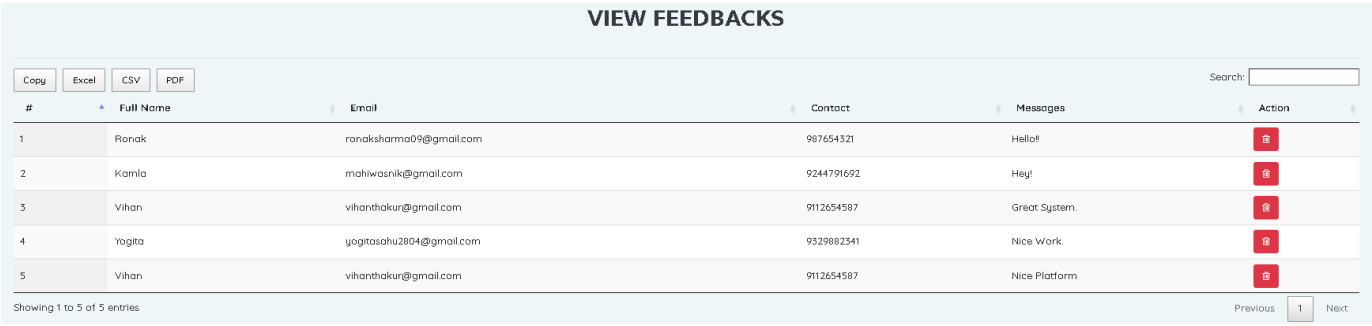
- The Doctor can view patient's prediction history.



Admin Dashboard-







- The Admin can view and manage doctor records, access patient details, and review feedback submitted by users through the admin dashboard.

12.System Maintenance & Evaluation

System Maintenance:

To ensure the Smart CardioCare system remains reliable, efficient, and up-to-date, proper maintenance practices are followed:

- Regular Updates: Machine learning models and software components are periodically updated to improve prediction accuracy and system performance.
- Bug Fixes: Any errors or glitches in the system are promptly identified and resolved.
- Database Maintenance: User data and system records are regularly backed up and optimized for efficient retrieval.
- Security Updates: Updates are applied to protect sensitive user data and prevent unauthorized access.

System Evaluation:

The system's performance and usability are evaluated to ensure it meets user needs and project objectives:

- Accuracy Testing: The machine learning model is tested with sample data to measure prediction accuracy (currently at 88.52%).
- Usability Testing: The interface is evaluated for user-friendliness, clarity, and ease of navigation.
- Performance Testing: The system's speed and responsiveness are checked under various scenarios to ensure smooth operation.
- Feedback and Improvement: User feedback is collected to identify areas for enhancement and implement improvements in future versions.

This maintenance and evaluation framework ensures that Smart CardioCare remains a reliable and effective tool for heart disease prediction and health monitoring.

13.Process Involved

The Smart CardioCare system follows a clear and structured process to predict heart disease risk and provide guidance to users:

1. **User Data Collection**:

   Users input their health parameters such as age, gender, blood pressure, cholesterol, heart rate, and other relevant medical data into the system.

2. **Data Validation**:

   The system verifies the entered data to ensure it is complete and within valid ranges.

3. **Data Processing**:

   The validated data is sent to the machine learning model, where it is analyzed using Gradient Boosting and Logistic Regression algorithms.

4. **Prediction Generation**:

   The model predicts the risk of heart disease based on the input data and outputs a result indicating low or high risk.

5. **Result Visualization**:

   The system presents the prediction result in an easy-to-understand format, including charts, risk scores, and recommendations.

6. **Guidance & Recommendations**:

   Based on the results, users receive suggestions for lifestyle improvements, preventive measures, or consultation with a doctor.

7. **Record Storage**:

   User data and prediction results are securely stored in the database for future reference and analysis.

This structured process ensures that users receive accurate, timely, and actionable insights into their heart health.

14.Testing

The Smart CardioCare system underwent several testing stages to ensure accuracy, usability, and reliability:

- **Unit Testing:**
  Each component, including user input forms and prediction modules, was tested individually to verify proper functionality.
- **Integration Testing:**
  The interaction between the front-end, back-end, and machine learning model was tested to ensure seamless data flow.
- **System Testing:**
  The complete system was evaluated to check end-to-end performance and stability.
- **Model Validation:**
  The machine learning algorithms were validated using test datasets, achieving an accuracy of approximately 88.52%.
- **User Testing:**
  Feedback from potential users was collected to assess ease of use, clarity of results, and overall experience.

This testing process guarantees that Smart CardioCare delivers accurate predictions and provides a seamless experience for users.

15. About Programming Languages

The Smart CardioCare system was developed using a combination of programming languages, frameworks, and libraries to ensure an efficient and reliable heart disease prediction platform:

Python: Used for implementing machine learning algorithms, data processing, and model training. Libraries such as NumPy, Pandas, and scikit-learn were employed to handle data manipulation and predictive analytics.

HTML, CSS, and JavaScript: Used for building the front-end interface, providing a clean, user-friendly, and interactive experience.

Bootstrap: Employed to make the web interface responsive and visually appealing across devices.

Django Framework: Used to connect the front-end with back-end processes, manage data flow, and handle user requests efficiently.

SQLite: Chosen as the database to store user information, prediction history, and system data due to its simplicity and speed.

These tools together enable a seamless integration between user input, prediction logic, and result presentation, making Smart CardioCare a robust and accessible platform for early heart disease detection.

16.About Database SQLITE

Database Used: SQLite

In the Smart CardioCare project, SQLite is used as the primary database to store and manage all application data. SQLite is a lightweight, file-based relational database that does not require a separate server, making it easy to integrate with Python and Django.

The database stores essential information such as:

- User health input data for heart disease prediction.
- System-generated prediction results and risk scores.
- Doctor details and recommendations.
- Other application-related data required for smooth operation.

Advantages of SQLite for this project:

- Lightweight and fast: Efficient for storing and retrieving data without heavy server requirements.
- Portable: The database file can easily move between different environments or devices.
- Easy integration: Works seamlessly with Python and Django frameworks.
- Reliable storage: Ensures that all user and system data is stored securely and can be accessed efficiently during predictions.

17.Print Out Code & Output

17.1 Database Models (models.py):-

```
from django.db import models
from django.contrib.auth.models import User
# Create your models here.
from.choices import DOCTOR_STATUS

class Patient(models.Model):
    user = models.ForeignKey(User, on_delete=models.CASCADE, null=True)
    contact = models.CharField(max_length=100, null=True)
    address = models.CharField(max_length=100, null=True)
    dob = models.DateField(null=True)
    image = models.FileField(null=True)

    def __str__(self):
        return self.user.username

class Doctor(models.Model):
    status = models.IntegerField(DOCTOR_STATUS, null=True)
    user = models.ForeignKey(User, on_delete=models.CASCADE, null=True)
    contact = models.CharField(max_length=100, null=True)
    address = models.CharField(max_length=100, null=True)
    category = models.CharField(max_length=100, null=True)
    doj = models.DateField(null=True)
    dob = models.DateField(null=True)
    image = models.FileField(null=True)
```

```
    def __str__(self):
        return self.user.username

class Admin_Health_CSV(models.Model):
    name = models.CharField(max_length=100, null=True)
    csv_file = models.FileField(null=True, blank=True)

    def __str__(self):
        return self.name

class Search_Data(models.Model):
    patient = models.ForeignKey(Patient, on_delete=models.CASCADE,
null=True)
    prediction_accuracy =
models.CharField(max_length=100,null=True,blank=True)
    result = models.CharField(max_length=100,null=True,blank=True)
    values_list = models.CharField(max_length=100,null=True,blank=True)
    created = models.DateTimeField(auto_now=True,null=True)

    def __str__(self):
        return self.patient.user.username

class Feedback(models.Model):
    user = models.ForeignKey(User, on_delete=models.CASCADE, null=True)
    messages = models.TextField(null=True)
    date = models.DateField(auto_now=True)
```

```
    def __str__(self):
        return self.user.username
```

17.2 Machine Learning Prediction Function(views.py):-

```
#This function trains the machine learning model and predicts heart disease risk
#based on user input
def prdict_heart_disease(list_data):

    csv_file = Admin_Helath_CSV.objects.get(id=1)
    df = pd.read_csv(csv_file.csv_file)

    X = df[['age','sex','cp','trestbps','chol','fbs',
            'restecg','thalach','exang','oldpeak',
            'slope','ca','thal']]

    y = df['target']

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, train_size=0.8, random_state=0)

    model = GradientBoostingClassifier(
        n_estimators=100,
        learning_rate=1.0,
        max_depth=1,
        random_state=0)
```

```
    model.fit(X_train, y_train)

    prediction = model.predict([list_data])

    accuracy = model.score(X_test, y_test) * 100

    return accuracy, prediction
```

17.3 Heart Disease Prediction View:-

```
@login_required(login_url="login")
def add_heartdetail(request):

    if request.method == "POST":

        list_data = []
        value_dict = eval(str(request.POST)[12:-1])

        count = 0
        for key, value in value_dict.items():

            if count == 0:
                count = 1
                continue

            if key == "sex" and value[0] in ["Male","male","M","m"]:
                list_data.append(0)
            elif key == "sex":
```

```
            list_data.append(1)
        else:
            list_data.append(value[0])

    accuracy, prediction = prdict_heart_disease(list_data)

    patient = Patient.objects.get(user=request.user)

    Search_Data.objects.create(
        patient=patient,
        prediction_accuracy=accuracy,
        result=prediction[0],
        values_list=list_data
    )

    return redirect('predict_desease',
                    str(prediction[0]),
                    str(accuracy))

  return render(request, 'add_heartdetail.html')
```

17.4 Prediction Result View:-

```
@login_required(login_url="login")
def predict_desease(request, pred, accuracy):

  doctors = Doctor.objects.all()
  context = {
```

```
        'pred': pred,
        'accuracy': accuracy,
        'doctor': doctors
    }

    return render(request,
                  'predict_disease.html',
                  context)
```

17.5User Registration View:-

```
def Signup_User(request):

    if request.method == 'POST':

        user = User.objects.create_user(
            username=request.POST['uname'],
            email=request.POST['email'],
            password=request.POST['pwd'],
            first_name=request.POST['fname'],
            last_name=request.POST['lname']
        )

        if request.POST['type'] == "Patient":

            Patient.objects.create(
                user=user,
                contact=request.POST['contact'],
```

```
            address=request.POST['add'],
            dob=request.POST['dob'],
            image=request.FILES['image']
        )

    else:

        Doctor.objects.create(
            user=user,
            contact=request.POST['contact'],
            address=request.POST['add'],
            dob=request.POST['dob'],
            image=request.FILES['image'],
            status=2
        )

    return render(request, 'register.html')
```

17.6 Doctor Dashboard View:-

```
@login_required(login_url="login")
def Doctor_Home(request):

    records = Search_Data.objects.all()

    total_patients = records.values('patient').distinct().count()

    healthy_patients = records.filter(
```

```
        result="0").values('patient').distinct().count()

    unhealthy_patients = records.filter(
        result="1").values('patient').distinct().count()

    context = {
        'total_patients': total_patients,
        'healthy_patients': healthy_patients,
        'unhealthy_patients': unhealthy_patients
    }

    return render(request,
                  'doctor_home.html',
                  context)
```

17.7 Key URL Patterns (urls.py):-

```
from django.urls import path
from health.views import *

urlpatterns = [
    path('', Home, name="home"),
    path('patient_home', User_Home, name="patient_home"),
    path('doctor_home', Doctor_Home, name="doctor_home"),
    path('admin_home', Admin_Home, name="admin_home"),
    path('add_heartdetail', add_heartdetail, name="add_heartdetail"),
    path('predict_desease/<str:pred>/<str:accuracy>/', predict_desease,
name="predict_desease"),
]
```

System Output Description:

The Smart CardioCare system successfully predicts heart disease risk based on user health inputs. After entering medical parameters, the system processes the data using the trained machine learning model and displays:

- Prediction Result (Healthy or At Risk)
- Prediction Accuracy Percentage
- Risk Visualization
- Doctor Recommendations

The system stores prediction history and displays statistics on the doctor dashboard.

18.Conclusion

The Smart CardioCare project successfully demonstrates the use of machine learning for heart disease prediction in a user-friendly web-based application. By allowing users to input their health parameters and receive instant risk predictions, the system empowers individuals to take proactive measures for their heart health.

The project integrates data-driven decision-making, visualization of risk factors, and guidance for lifestyle and medical consultation, making it a comprehensive tool for early detection and prevention. The model achieved an accuracy of 88.52%, ensuring reliable results for practical use.

Overall, the system provides a robust, accessible, and efficient solution to support healthcare awareness and preventive measures for cardiovascular diseases.

19.Future Aspects

- Integrate wearable devices for real-time monitoring
- Develop a mobile app for easier access
- Improve prediction accuracy with advanced AI models
- Provide personalized health insights
- Connect with telemedicine platforms
- Expand to predict other chronic diseases

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